On the Ambrosetti–Brezis–Cerami conjecture

Anderson L.A. de Araujo* Luiz F.O. Faria†

Abstract

We revisit the small-parameter uniqueness problem for the concave–convex Dirichlet equation

$$-\Delta u = \lambda u^q + u^p \quad \text{in } B_1, \qquad u > 0 \quad \text{in } B_1, \qquad u = 0 \quad \text{on } \partial B_1,$$

where $B_1 \subset \mathbb{R}^N$, $N \geq 3$, $0 < q < 1$, and $p > (N+2)/(N-2)$. This problem originates from an open question of Ambrosetti, Brezis and Cerami concerning multiplicity in the Sobolev-supercritical regime, for which small-parameter uniqueness was subsequently stated by Zhao and Zhong through an approach based on radial reduction and singular solutions. We give a new, self-contained proof based on the radial first-zero map. Small amplitudes are controlled by the pure concave limit, intermediate amplitudes by a uniform lower bound for the first zero, and large amplitudes by an Emden–Fowler analysis of the supercritical Lane–Emden equation. These estimates yield a unique positive radial solution for all sufficiently small $\lambda > 0$. A symmetry argument adapted to the non-Lipschitz concave term then gives uniqueness among all positive classical solutions. Moreover,

$$\|u_\lambda\|_{L^\infty(B_1)} \sim \rho_q^{-\frac{2}{1-q}} \lambda^{\frac{1}{1-q}} \qquad \text{as } \lambda \to 0^+,$$

where ρ_q is the first positive zero of the normalized pure concave profile.

1 Introduction and main result

Let

$$B_1 := B_1(0) \subset \mathbb{R}^N, \qquad N \geq 3,$$

and consider the concave–convex Dirichlet problem

$$\begin{cases} -\Delta u = \lambda u^q + u^p, & \text{in } B_1, \\ u > 0, & \text{in } B_1, \\ u = 0, & \text{on } \partial B_1, \end{cases} \tag{1}$$

where

$$0 < q < 1, \qquad p > p_c := \frac{N+2}{N-2}, \qquad \lambda > 0. \tag{2}$$

The study of (1) goes back to the classical work of Ambrosetti, Brezis and Cerami [1], where the combined effect of a concave and a convex nonlinearity was shown to produce a rich existence and multiplicity theory. In particular, they proved the existence of a minimal positive solution

*Universidade Federal de Viçosa, Departamento de Matemática, Avenida Peter Henry Rolfs, Viçosa, MG, Brazil, CEP 36570-000. E-mail: anderson.araujo@ufv.br

†Universidade Federal de Juiz de Fora, Departamento de Matemática, Rua José Lourenço Kelmer, s/n, Juiz de Fora, MG, Brazil, CEP 36036-900. E-mail: luiz.faria@ufjf.br

for sufficiently small $\lambda > 0$ and obtained a second positive solution in the Sobolev-subcritical and critical range

$$1 < p \leq \frac{N+2}{N-2}.$$

Their analysis left open the corresponding multiplicity question in the Sobolev-supercritical regime: if

$$p > \frac{N+2}{N-2}$$

and the domain is a ball, does a second positive solution still exist for sufficiently small $\lambda > 0$?

The supercritical case is substantially different from the subcritical and critical ones. The usual compact variational framework is no longer available, and the possible behavior of large-amplitude solutions becomes an essential part of the problem. A radial multiplicity result was subsequently stated by Ma [3] for $0 < q < 1 < p$, with a formulation including the supercritical range. On the other hand, Zhao and Zhong [6] later reached the opposite conclusion for the problem considered here: for

$$0 < q < 1, \qquad p > \frac{N+2}{N-2},$$

they stated uniqueness of the positive solution for sufficiently small $\lambda > 0$, thus giving a negative answer to the open question of Ambrosetti, Brezis and Cerami.

The argument of Zhao–Zhong is based on the asymptotic behavior of large solutions and on a distinguished singular radial solution. After reducing the problem to the radial equation, they associate with the singular solution a parameter $\lambda_* > 0$ and assert that every sequence of regular solutions (λ_n, u_n) satisfying

$$\|u_n\|_{L^\infty(B_1)} \longrightarrow \infty$$

must satisfy

$$\lambda_n \longrightarrow \lambda_* > 0,$$

with u_n converging to the singular profile away from the origin. Their small-parameter uniqueness result is then obtained by combining this asymptotic statement with the assertion that any solution different from the minimal one must satisfy

$$\|v_\lambda\|_{L^\infty(B_1)} \longrightarrow \infty \qquad \text{as } \lambda \to 0^+.$$

Some steps in the published proof of this argument are presented only in abbreviated form. In particular, the latter blow-up alternative, which is essential in the deduction of small-parameter uniqueness, is stated in [6] without proof, with reference to an argument from the earlier Ambrosetti–Brezis–Cerami theory. Moreover, the asymptotic identification of an arbitrary unbounded sequence with the distinguished singular orbit involves limiting steps that are only briefly justified in the published proof. We do not claim that these points cannot be completed, and the purpose of the present work is not to provide a corrigendum to [6]. Instead, we develop a different, self-contained approach which avoids the singular-solution limiting argument altogether.

The main idea is to study directly the geometry of the first-zero map associated with the radial shooting problem. After an appropriate scaling, problem (1) is transformed into

$$\begin{cases} -\Delta U = U^q + U^p, & \text{in } B_R, \\ U > 0, & \text{in } B_R, \\ U = 0, & \text{on } \partial B_R, \end{cases} \tag{3}$$

where

$$R = \lambda^{\frac{p-1}{2(p-q)}}. \tag{4}$$

Thus the limit $\lambda \to 0^+$ corresponds precisely to the small-radius limit $R \to 0^+$.

For $A > 0$, let U_A denote the radial solution of

$$U_A'' + \frac{N-1}{r}U_A' + U_A^q + U_A^p = 0, \qquad U_A(0) = A, \qquad U_A'(0) = 0, \tag{5}$$

and let $R(A)$ be its first positive zero. Positive radial solutions of (3) are therefore characterized by

$$R(A) = R.$$

The uniqueness problem is reduced to understanding the first-zero map $A \mapsto R(A)$ when R is small.

Our analysis separates naturally into three amplitude regimes. For small A, the concave term dominates. With the rescaling

$$U_A(r) = A\, V_A\Big(A^{-\frac{1-q}{2}} r\Big),$$

the functions V_A converge to the normalized solution V of

$$V'' + \frac{N-1}{r}V' + V^q = 0, \qquad V(0) = 1, \qquad V'(0) = 0. \tag{6}$$

If ρ_q denotes the first positive zero of V, we obtain

$$R(A) = \rho_q A^{\frac{1-q}{2}}(1 + o(1)) \qquad \text{as } A \to 0^+. \tag{7}$$

In addition, a Picone-type argument gives uniqueness below the explicit amplitude

$$A_* = \left(\frac{1-q}{p-1}\right)^{\frac{1}{p-q}}, \tag{8}$$

which is exactly the interval on which

$$s \longmapsto \frac{s^q + s^p}{s} = s^{q-1} + s^{p-1}$$

is strictly decreasing.

The essential issue is therefore to rule out additional solutions whose central values stay away from zero. On every fixed compact amplitude interval

$$0 < A_0 \le A \le A_1 < \infty,$$

an elementary shooting estimate provides a uniform positive lower bound for $R(A)$. The genuinely supercritical mechanism appears in the large-amplitude regime. Setting

$$U_A(r) = A\, v_\varepsilon\left(A^{\frac{p-1}{2}} r\right), \qquad \varepsilon = A^{q-p},$$

leads to

$$v_\varepsilon'' + \frac{N-1}{r}v_\varepsilon' + v_\varepsilon^p + \varepsilon v_\varepsilon^q = 0.$$

As $A \to \infty$, equivalently $\varepsilon \to 0^+$, this equation approaches the supercritical Lane–Emden equation

$$v'' + \frac{N-1}{r}v' + v^p = 0.$$

The Emden–Fowler transformation converts the limiting equation into an autonomous planar system possessing a positive equilibrium. The condition

$$p > \frac{N+2}{N-2}$$

is precisely what makes the damping coefficient positive. The equilibrium is consequently asymptotically stable, and a robust trapping argument shows that the small non-autonomous perturbation generated by the concave term cannot drive the corresponding orbit to zero before a sufficiently long Emden–Fowler time. After returning to the original variables, the scaling factors cancel exactly and yield a positive lower bound for the first zero of all large-amplitude profiles.

Combining the intermediate- and large-amplitude estimates, we prove the fundamental separation property

$$\inf_{A \geq A_0} R(A) > 0 \qquad \text{for every } A_0 > 0. \tag{9}$$

Hence, if $R > 0$ is sufficiently small, every positive radial solution of (3) necessarily belongs to the small-amplitude branch, where uniqueness is already available. In particular, the argument excludes a second solution directly at the level of the shooting diagram; no passage to a singular limiting solution and no a priori blow-up alternative for a hypothetical second branch are required.

This gives a different dynamical explanation of the small-parameter phenomenon in the supercritical regime. The contribution of the present work is therefore not only a new proof of the uniqueness statement. The first-zero analysis provides a self-contained mechanism that controls the complete amplitude range, makes explicit where the supercritical assumption enters the argument, and yields quantitative information on the unique small branch. In particular, combining (4) and (7), we obtain

$$\|u_\lambda\|_{L^\infty(B_1)} \sim \rho_q^{-\frac{2}{1-q}} \lambda^{\frac{1}{1-q}} \qquad \text{as } \lambda \to 0^+. \tag{10}$$

There remains one point that requires some care. Since $0 < q < 1$, the reaction

$$f_\lambda(s) = \lambda s^q + s^p$$

is not locally Lipschitz at $s = 0$. Thus the passage from radial uniqueness to uniqueness among all positive solutions should not be based without further justification on a moving-plane theorem whose standard formulation assumes local Lipschitz continuity of the reaction. We use instead a symmetry result applicable to the present structure, observing that

$$f_\lambda(s) = (s^+)^p + \lambda (s^+)^q$$

is the sum of a locally Lipschitz function and a nondecreasing function. It follows that every positive classical solution of (1) in the ball is radial and strictly decreasing in the radial variable. Consequently, the radial uniqueness obtained from the shooting analysis yields uniqueness among all positive classical solutions.

Our main result can now be stated as follows.

Theorem 1.1. *Assume* (2). *Then there exists* $\lambda_* > 0$ *such that, for every*

$$0 < \lambda < \lambda_*,$$

problem (1) *admits exactly one positive classical solution, denoted by* u_λ. *Moreover,* u_λ *is radial and, writing*

$$u_\lambda(x) = U_\lambda(|x|),$$

one has

$$U_\lambda'(r) < 0 \qquad \textit{for every } 0 < r < 1.$$

Furthermore,

$$\|u_\lambda\|_{L^\infty(B_1)} \sim \rho_q^{-\frac{2}{1-q}} \lambda^{\frac{1}{1-q}} \qquad \textit{as } \lambda \to 0^+,$$

where ρ_q denotes the first positive zero of the solution of

$$V'' + \frac{N-1}{r} V' + V^q = 0, \qquad V(0) = 1, \qquad V'(0) = 0.$$

The proof of Theorem 1.1 is organized around four main ingredients. First, we perform the scaling that transforms the small-parameter problem into the parameter-free equation on a small ball. Second, we analyze the pure concave limit and establish uniqueness below the explicit amplitude threshold (8). Third, we study the first-zero map over the intermediate and large-amplitude regimes, the latter through the Emden–Fowler dynamics of the supercritical Lane–Emden equation. Finally, we apply the appropriate symmetry principle for the non-Lipschitz concave–convex reaction to pass from radial uniqueness to full uniqueness.

2 Scaling and the shooting formulation

Set

$$\gamma := \frac{p-1}{2(p-q)}. \tag{11}$$

For $\lambda > 0$, consider the change of variables

$$u(x) = \lambda^{\frac{1}{p-q}} U(\lambda^\gamma x)\,. \tag{12}$$

Then, under this change of variables, u solves (1) if and only if U solves

$$\begin{cases} -\Delta U = U^q + U^p, & \text{in } B_R, \\ U > 0, & \text{in } B_R, \\ U = 0, & \text{on } \partial B_R, \end{cases} \tag{13}$$

where

$$R = \lambda^\gamma. \tag{14}$$

Indeed,

$$-\Delta_x u = \lambda^{\frac{1}{p-q}+2\gamma}(-\Delta U) = \lambda^{\frac{p}{p-q}}(-\Delta U),$$

while

$$\lambda u^q = \lambda^{1+\frac{q}{p-q}} U^q = \lambda^{\frac{p}{p-q}} U^q$$

and

$$u^p = \lambda^{\frac{p}{p-q}} U^p.$$

Thus the small-λ regime is equivalent to the small-radius regime

$$R \to 0^+.$$

For $A > 0$, let U_A denote the maximal positive solution of the radial initial-value problem

$$U_A'' + \frac{N-1}{r} U_A' + U_A^q + U_A^p = 0, \qquad U_A(0) = A, \qquad U_A'(0) = 0. \tag{15}$$

Define the endpoint of its interval of positivity by

$$R(A) := \inf\{r > 0 : U_A(r) = 0\}, \tag{16}$$

with the convention $R(A) = +\infty$ if U_A remains positive for all $r > 0$.

For every $0 < r < R(A)$, integration of (15) gives

$$-r^{N-1}U_A'(r) = \int_0^r s^{N-1}\left(U_A(s)^q + U_A(s)^p\right)\, ds > 0. \tag{17}$$

Hence every positive shooting profile is strictly decreasing on $(0, R(A))$. Consequently, positive radial solutions of (13) are in one-to-one correspondence with values $A > 0$ satisfying

$$R(A) = R.$$

3 The small-amplitude branch

Define

$$A_* := \left(\frac{1-q}{p-1}\right)^{\frac{1}{p-q}}. \tag{18}$$

Lemma 3.1. *The function*

$$s \longmapsto \frac{s^q + s^p}{s} = s^{q-1} + s^{p-1}$$

is strictly decreasing on $(0, A_*)$.

Proof. A direct computation gives

$$\frac{d}{ds}\left(s^{q-1} + s^{p-1}\right) = s^{q-2}\left[-(1-q) + (p-1)s^{p-q}\right].$$

The expression in brackets is negative exactly when

$$s^{p-q} < \frac{1-q}{p-1},$$

that is, for $0 < s < A_*$. □

Lemma 3.2. *Let* $\Omega \subset \mathbb{R}^N$ *be a bounded domain and let* $I = (0, M)$ *for some* $M > 0$. *Assume that*

$$f : I \longrightarrow (0, \infty)$$

is continuous and that

$$s \longmapsto \frac{f(s)}{s}$$

is strictly decreasing on I.

Suppose that $u, v \in H_0^1(\Omega) \cap C(\overline{\Omega})$ *are positive weak solutions of*

$$-\Delta w = f(w) \qquad \text{in } \Omega,$$

and that

$$0 < u(x), v(x) < M \qquad \text{for every } x \in \Omega.$$

Assume moreover that

$$f(u)u,\ f(v)v \in L^1(\Omega).$$

Then

$$u \equiv v.$$

Proof. For $\varepsilon > 0$, Picone's identity applied to u and $v + \varepsilon$ gives

$$|\nabla u|^2 - \nabla(v+\varepsilon) \cdot \nabla \left(\frac{u^2}{v+\varepsilon} \right) = \left| \nabla u - \frac{u}{v+\varepsilon} \nabla v \right|^2 \geq 0.$$

Since

$$\nabla(v+\varepsilon) = \nabla v,$$

integration yields

$$\int_\Omega |\nabla u|^2 \, dx \geq \int_\Omega \nabla v \cdot \nabla \left(\frac{u^2}{v+\varepsilon} \right) dx. \tag{19}$$

The function

$$\frac{u^2}{v+\varepsilon}$$

belongs to $H_0^1(\Omega)$ for every fixed $\varepsilon > 0$, and therefore it is an admissible test function in the weak equation for v. Consequently,

$$\int_\Omega \nabla v \cdot \nabla \left(\frac{u^2}{v+\varepsilon} \right) dx = \int_\Omega f(v) \frac{u^2}{v+\varepsilon} \, dx.$$

Testing the equation for u with u gives

$$\int_\Omega |\nabla u|^2 \, dx = \int_\Omega f(u) u \, dx.$$

Thus

$$\int_\Omega f(u) u \, dx \geq \int_\Omega f(v) \frac{u^2}{v+\varepsilon} \, dx. \tag{20}$$

Since the integrand on the right-hand side is nonnegative and

$$f(v) \frac{u^2}{v+\varepsilon} \longrightarrow \frac{f(v)}{v} u^2 \qquad \text{a.e. in } \Omega$$

as $\varepsilon \to 0^+$, Fatou's lemma gives

$$\int_\Omega \frac{f(v)}{v} u^2 \, dx \leq \liminf_{\varepsilon \to 0^+} \int_\Omega f(v) \frac{u^2}{v+\varepsilon} \, dx \leq \int_\Omega f(u) u \, dx.$$

Hence

$$\int_\Omega \left[\frac{f(u)}{u} - \frac{f(v)}{v} \right] u^2 \, dx \geq 0. \tag{21}$$

Interchanging u and v, we likewise obtain

$$\int_\Omega \left[\frac{f(v)}{v} - \frac{f(u)}{u} \right] v^2 \, dx \geq 0. \tag{22}$$

Adding (21) and (22), we obtain

$$\int_\Omega \left[\frac{f(u)}{u} - \frac{f(v)}{v} \right] (u^2 - v^2) \, dx \geq 0. \tag{23}$$

On the other hand, since $f(s)/s$ is strictly decreasing,

$$\left[\frac{f(u)}{u} - \frac{f(v)}{v}\right](u - v) \leq 0 \qquad \text{a.e. in } \Omega,$$

with strict inequality wherever $u \neq v$. Since

$$u^2 - v^2 = (u - v)(u + v)$$

and $u + v > 0$, it follows that

$$\left[\frac{f(u)}{u} - \frac{f(v)}{v}\right](u^2 - v^2) \leq 0 \qquad \text{a.e. in } \Omega,$$

with strict inequality wherever $u \neq v$.

Together with (23), this forces

$$u = v \qquad \text{a.e. in } \Omega.$$

By continuity,

$$u \equiv v \qquad \text{in } \overline{\Omega}.$$

□

Proposition 3.3. *For every $R > 0$, problem* (13) *has at most one positive solution U satisfying*

$$\|U\|_{L^\infty(B_R)} < A_*.$$

Proof. Let U_1, U_2 be two such solutions and set

$$f(s) := s^q + s^p.$$

By Lemma 3.1, the map

$$s \longmapsto \frac{f(s)}{s}$$

is strictly decreasing on the range of both solutions.

Moreover,

$$0 < U_i(x) < A_* \qquad (x \in B_R,\ i = 1, 2),$$

and

$$f(U_i)U_i = U_i^{q+1} + U_i^{p+1} \in L^1(B_R),$$

since U_i is bounded. Therefore all the hypotheses of Lemma 3.2 are satisfied with

$$\Omega = B_R, \qquad M = A_*.$$

It follows directly that

$$U_1 \equiv U_2.$$

□

We next identify the first-zero behavior as $A \to 0^+$.

Lemma 3.4. *Assume that $0 < q < 1$. Let V be the positive solution, up to its first zero, of*

$$V'' + \frac{N-1}{s}V' + V^q = 0, \qquad V(0) = 1, \qquad V'(0) = 0. \tag{24}$$

If ρ_q denotes its first positive zero, then

$$0 < \rho_q \leq \sqrt{\frac{2N}{1-q}} < \infty, \qquad V'(\rho_q) < 0.$$

In particular, the first zero is transversal.

Proof. As long as $V > 0$, equation (24) can be written as

$$\left(s^{N-1}V'(s)\right)' = -s^{N-1}V(s)^q.$$

Using $V'(0) = 0$, we obtain

$$-V'(s) = s^{1-N}\int_0^s t^{N-1}V(t)^q\,dt > 0.$$

Thus V is strictly decreasing on its interval of positivity. Consequently, for

$$0 \leq t \leq s,$$

$$V(t) \geq V(s),$$

and hence

$$\begin{aligned} -V'(s) &= s^{1-N}\int_0^s t^{N-1}V(t)^q\,dt \\ &\geq s^{1-N}V(s)^q\int_0^s t^{N-1}\,dt = \frac{s}{N}V(s)^q. \end{aligned}$$

Therefore

$$-\frac{V'(s)}{V(s)^q} \geq \frac{s}{N}.$$

Since $0 < q < 1$,

$$\frac{d}{ds}V(s)^{1-q} = (1-q)V(s)^{-q}V'(s),$$

and hence

$$\frac{d}{ds}V(s)^{1-q} \leq -\frac{1-q}{N}s.$$

Integrating from 0 to s and using $V(0) = 1$ gives

$$V(s)^{1-q} \leq 1 - \frac{1-q}{2N}s^2.$$

If V remained positive up to

$$s_* := \sqrt{\frac{2N}{1-q}},$$

then the preceding inequality would give

$$0 < V(s_*)^{1-q} \leq 0,$$

which is impossible. Therefore V has a first positive zero ρ_q satisfying

$$0 < \rho_q \le \sqrt{\frac{2N}{1-q}} < \infty.$$

Finally, passing to the limit $s \uparrow \rho_q$ in the integral identity gives

$$-\rho_q^{N-1} V'(\rho_q) = \int_0^{\rho_q} t^{N-1} V(t)^q \, dt.$$

The integral on the right-hand side is strictly positive, since $V(t) > 0$ for $0 \le t < \rho_q$. Hence

$$V'(\rho_q) < 0.$$

Thus the first zero is transversal. □

Proposition 3.5. *Assume that $0 < q < 1 < p$. Then, as $A \to 0^+$,*

$$R(A) = \rho_q A^{\frac{1-q}{2}} (1 + o(1)). \tag{25}$$

Proof. Set

$$U_A(r) = A V_A(s), \qquad s = A^{-\frac{1-q}{2}} r.$$

Then V_A solves

$$V_A'' + \frac{N-1}{s} V_A' + V_A^q + A^{p-q} V_A^p = 0, \qquad V_A(0) = 1, \qquad V_A'(0) = 0. \tag{26}$$

Let ρ_A denote the first positive zero of V_A.

We first prove

$$\liminf_{A \to 0^+} \rho_A \ge \rho_q.$$

Fix $\delta \in (0, \rho_q)$ and set

$$m_\delta := \min_{0 \le s \le \rho_q - \delta} V(s) > 0.$$

On a compact neighborhood of $[m_\delta, 1]$ contained in $(0, \infty)$, the maps $t \mapsto t^q$ and $t \mapsto t^p$ are locally Lipschitz. Since $A^{p-q} \to 0$, continuous dependence for the radial initial-value problem, equivalently from its integral formulation, yields

$$V_A \longrightarrow V \qquad \text{in } C^1([0, \rho_q - \delta])$$

as $A \to 0^+$. Hence, for all sufficiently small A,

$$V_A(s) \ge \frac{m_\delta}{2} > 0 \qquad (0 \le s \le \rho_q - \delta),$$

and therefore

$$\rho_A > \rho_q - \delta.$$

Since $\delta > 0$ is arbitrary,

$$\liminf_{A \to 0^+} \rho_A \ge \rho_q.$$

We next prove the reverse inequality. Since

$$V(\rho_q) = 0 \qquad \text{and} \qquad V'(\rho_q) < 0,$$

choose $c_0 > 0$ and $\eta > 0$ such that

$$V'(s) \le -2c_0 \qquad \text{for } s \in [\rho_q - \eta, \rho_q].$$

Fix $\varepsilon > 0$. Choose

$$s_0 \in (\rho_q - \min\{\eta, \varepsilon/2\}, \rho_q)$$

so close to ρ_q that

$$V(s_0) < \frac{c_0\varepsilon}{4}\left(\frac{s_0}{s_0 + \varepsilon/2}\right)^{N-1}.$$

By continuous dependence on the fixed interval $[0, s_0]$,

$$V_A(s_0) \to V(s_0), \qquad V_A'(s_0) \to V'(s_0).$$

Thus, for all sufficiently small A,

$$V_A'(s_0) \le -c_0$$

and

$$V_A(s_0) < \frac{c_0\varepsilon}{2}\left(\frac{s_0}{s_0 + \varepsilon/2}\right)^{N-1}.$$

As long as V_A remains positive, equation (26) implies

$$\left(s^{N-1}V_A'(s)\right)' = -s^{N-1}\left(V_A(s)^q + A^{p-q}V_A(s)^p\right) < 0.$$

Therefore, for $s \ge s_0$, as long as $V_A(s) > 0$,

$$s^{N-1}V_A'(s) \le s_0^{N-1}V_A'(s_0),$$

and hence

$$V_A'(s) \le -c_0\left(\frac{s_0}{s}\right)^{N-1}.$$

Suppose, by contradiction, that V_A remains positive on $[s_0, s_0 + \varepsilon/2]$. Then

$$V_A'(s) \le -c_0\left(\frac{s_0}{s_0 + \varepsilon/2}\right)^{N-1}$$

throughout that interval. Hence

$$\begin{aligned} V_A(s_0 + \varepsilon/2) &\le V_A(s_0) - \frac{c_0\varepsilon}{2}\left(\frac{s_0}{s_0 + \varepsilon/2}\right)^{N-1} \\ &< 0, \end{aligned}$$

a contradiction. Therefore

$$\rho_A < s_0 + \frac{\varepsilon}{2} < \rho_q + \varepsilon.$$

It follows that

$$\limsup_{A \to 0^+} \rho_A \le \rho_q.$$

Combining the two inequalities, we conclude that

$$\rho_A \longrightarrow \rho_q.$$

Returning to the original radial variable,

$$R(A) = A^{\frac{1-q}{2}}\rho_A,$$

and therefore

$$R(A) = \rho_q A^{\frac{1-q}{2}}(1 + o(1)),$$

which proves (25). □

Corollary 3.6. *There exist $R_0 > 0$ and $A_0 \in (0, A_*)$ such that, for every $R \in (0, R_0)$, problem (13) admits a positive radial solution U satisfying*

$$U(0) < A_0.$$

Moreover,

$$\|U\|_{L^\infty(B_R)} = U(0) < A_0 < A_*.$$

Proposition 3.3 applies to all positive solutions of (13), then this solution is unique among all positive solutions satisfying

$$\|U\|_{L^\infty(B_R)} < A_*.$$

Proof. By Proposition 3.5,

$$R(A) = \rho_q A^{\frac{1-q}{2}}(1 + o(1)) \qquad \text{as } A \to 0^+.$$

In particular,

$$R(A) \to 0 \qquad \text{as } A \to 0^+.$$

The transversality of the first zero, together with continuous dependence of the radial initial-value problem on the initial value, implies persistence and continuous dependence of the first zero. Hence there exists $\delta > 0$ such that

$$A \longmapsto R(A)$$

is continuous on $(0, \delta)$.

Choose

$$A_0 \in (0, \min\{\delta, A_*\})$$

and set

$$R_0 := R(A_0) > 0.$$

Fix $R \in (0, R_0)$. Since

$$R(A) \to 0 \qquad \text{as } A \to 0^+,$$

there exists $A_1 \in (0, A_0)$ such that

$$R(A_1) < R.$$

On the other hand,

$$R(A_0) = R_0 > R.$$

By continuity of the first-zero map, there exists

$$A \in (A_1, A_0)$$

such that

$$R(A) = R.$$

The corresponding shooting profile U_A is therefore a positive radial solution of (13) in B_R.

Moreover, every positive radial shooting profile is strictly decreasing on its interval of positivity. Hence

$$\|U_A\|_{L^\infty(B_R)} = U_A(0) = A < A_0 < A_*.$$

Finally, if Proposition 3.3 gives uniqueness among all positive solutions satisfying

$$\|U\|_{L^\infty(B_R)} < A_*,$$

then the asserted uniqueness follows directly. □

4 The supercritical Lane–Emden orbit

We now study the radial initial-value problem

$$v_0'' + \frac{N-1}{r} v_0' + v_0^p = 0, \qquad v_0(0) = 1, \qquad v_0'(0) = 0. \tag{27}$$

Set

$$a := \frac{2}{p-1}, \qquad A := N - 2 - 2a, \qquad B := a(N-2-a). \tag{28}$$

Since $p > p_c$, we have

$$0 < a < \frac{N-2}{2},$$

and therefore

$$A > 0, \qquad B > 0.$$

Introduce the Emden–Fowler variables

$$t = \log r, \qquad x_0(t) = r^a v_0(r).$$

Then x_0 satisfies

$$x_0'' + A x_0' - B x_0 + x_0^p = 0. \tag{29}$$

The unique positive equilibrium of (29) is

$$\ell := B^{\frac{1}{p-1}}. \tag{30}$$

Lemma 4.1. *Assume that $N \geq 3$ and*

$$p > p_c := \frac{N+2}{N-2}.$$

The solution v_0 of (27) is positive on $(0, \infty)$ and

$$x_0(t) \longrightarrow \ell, \qquad x_0'(t) \longrightarrow 0 \qquad \text{as } t \to +\infty.$$

Equivalently,

$$r^{\frac{2}{p-1}} v_0(r) \longrightarrow \left[\frac{2}{p-1} \left(N - 2 - \frac{2}{p-1} \right) \right]^{\frac{1}{p-1}} \qquad \text{as } r \to +\infty. \tag{31}$$

Proof. Suppose that v_0 had a first zero at some finite radius $R > 0$. Since $v_0 > 0$ in $(0, R)$, it would define a nontrivial positive solution of

$$-\Delta v = v^p \quad \text{in } B_R, \qquad v = 0 \quad \text{on } \partial B_R.$$

Since $p > p_c$ and B_R is star-shaped, the Pohozaev identity excludes the existence of such a solution. Hence

$$v_0(r) > 0 \qquad \text{for every } r > 0.$$

For (29), consider the Lyapunov function

$$E(x, y) := \frac{1}{2}y^2 - \frac{B}{2}x^2 + \frac{1}{p+1}x^{p+1}.$$

Along the orbit (x_0, x_0'),

$$\frac{d}{dt}E(x_0(t), x_0'(t)) = -A(x_0'(t))^2 \le 0.$$

Regularity of the radial solution at the origin gives

$$v_0(r) = 1 + O(r^2), \qquad v_0'(r) = O(r) \qquad \text{as } r \to 0^+.$$

Consequently,

$$x_0(t) = e^{at}(1 + o(1)), \qquad x_0'(t) = ae^{at}(1 + o(1)) \qquad \text{as } t \to -\infty.$$

Consequently,

$$E(x_0(t), x_0'(t)) \longrightarrow 0 \qquad \text{as } t \to -\infty.$$

Moreover,

$$E(x_0(t), x_0'(t)) = -A\int_{-\infty}^{t} (x_0'(s))^2\, ds.$$

Since

$$x_0'(s) \sim ae^{as} > 0 \qquad \text{as } s \to -\infty,$$

the integral on the right-hand side is strictly positive for every finite t. It follows that

$$E(x_0(t), x_0'(t)) < 0 \qquad \text{for every finite } t.$$

We next show that the positive semi-orbit is bounded. Since

$$E(x_0, x_0') = \frac{1}{2}(x_0')^2 - \frac{B}{2}x_0^2 + \frac{1}{p+1}x_0^{p+1} < 0,$$

we obtain

$$\frac{1}{p+1}x_0^{p+1} < \frac{B}{2}x_0^2.$$

Since $x_0 > 0$, this yields the explicit bound

$$0 < x_0(t) < \left(\frac{B(p+1)}{2}\right)^{\frac{1}{p-1}} \qquad \text{for every } t \in \mathbb{R}.$$

Thus x_0 is uniformly bounded.

Moreover, again using $E < 0$,

$$\frac{1}{2}(x_0')^2 < \frac{B}{2}x_0^2 - \frac{1}{p+1}x_0^{p+1} \le \frac{B}{2}x_0^2,$$

and therefore

$$|x_0'(t)| < \sqrt{B}\, x_0(t).$$

Hence x_0' is uniformly bounded as well, and the positive semi-orbit is relatively compact in the phase plane.

Since E is nonincreasing and bounded below along the orbit, there exists

$$E_\infty := \lim_{t\to+\infty} E(x_0(t), x_0'(t)).$$

For every fixed $t_0 \in \mathbb{R}$,

$$E_\infty \le E(x_0(t_0), x_0'(t_0)) < 0,$$

so

$$E_\infty < 0.$$

To apply LaSalle's invariance principle without any difficulty at $x = 0$ when p is not an integer, extend the vector field to the whole phase plane by replacing x^p with $|x|^{p-1}x$. Thus consider

$$x' = y, \qquad y' = -Ay + Bx - |x|^{p-1}x.$$

This system agrees with (29) along the positive orbit. The corresponding globally defined Lyapunov function is

$$\widetilde{E}(x,y) = \frac{1}{2}y^2 - \frac{B}{2}x^2 + \frac{1}{p+1}|x|^{p+1},$$

and along its trajectories

$$\frac{d}{dt}\widetilde{E}(x(t), y(t)) = -Ay(t)^2.$$

Because the positive semi-orbit is relatively compact, its ω-limit set is nonempty, compact, connected, and invariant. By LaSalle's invariance principle it is contained in the largest invariant subset of

$$\{(x,y) : y = 0\}.$$

Since the ω-limit set is contained in the nonnegative half-plane, invariance then forces

$$Bx - x^p = 0.$$

Thus the only nonnegative invariant points in this set are the equilibria

$$(0,0) \qquad \text{and} \qquad (\ell, 0).$$

The equilibrium $(0,0)$ has energy 0, whereas the energy is constant and equal to $E_\infty < 0$ on the ω-limit set. Hence $(0,0)$ cannot belong to it. Since the ω-limit set is nonempty, we conclude that

$$\omega(x_0, x_0') = \{(\ell, 0)\}.$$

Consequently,

$$x_0(t) \longrightarrow \ell, \qquad x_0'(t) \longrightarrow 0 \qquad \text{as } t \to +\infty.$$

Finally, since

$$\ell = B^{\frac{1}{p-1}} = \left[\frac{2}{p-1}\left(N - 2 - \frac{2}{p-1}\right)\right]^{\frac{1}{p-1}},$$

the preceding convergence is equivalent to (31). This proves the assertion. □

The equilibrium $(\ell, 0)$ is exponentially asymptotically stable. Indeed, writing $x = \ell + z$ and linearizing (29) at $(\ell, 0)$ gives

$$z'' + Az' + (p-1)Bz = 0.$$

Its characteristic polynomial is

$$\mu^2 + A\mu + (p-1)B.$$

Since

$$A > 0 \qquad \text{and} \qquad (p-1)B > 0,$$

both characteristic roots have strictly negative real parts. Hence the corresponding first-order linearized matrix is Hurwitz, and $(\ell, 0)$ is a hyperbolic exponentially asymptotically stable equilibrium of the autonomous Emden–Fowler system.

We record the following robust stability property, which will be used in the large-amplitude analysis.

Lemma 4.2. *There exist numbers $\eta > 0$, $\eta_0 > 0$, and $\delta > 0$ with*

$$0 < \eta < \frac{\ell}{4}$$

such that the following holds.

Let $d : [T, T_1] \to [0, \delta]$ be continuous and let (x, y) be a solution, as long as $x > 0$, of

$$\begin{cases} x' = y, \\ y' = -Ay + Bx - x^p - d(t)x^q, \end{cases} \tag{32}$$

with

$$|x(T) - \ell| + |y(T)| < \eta_0.$$

Then

$$|x(t) - \ell| + |y(t)| < \eta \qquad \text{for every } t \in [T, T_1].$$

In particular,

$$x(t) > \frac{\ell}{2} \qquad (T \le t \le T_1).$$

Proof. Set

$$Z = \begin{pmatrix} x - \ell \\ y \end{pmatrix}.$$

In what follows we use the ℓ^1-norm

$$|Z| := |x - \ell| + |y|,$$

together with its induced matrix norm. Since $\ell > 0$, we shall choose $\eta < \ell/4$. Thus, whenever $|Z| \le \eta$, one has

$$x > \ell - \eta > \frac{3\ell}{4} > 0,$$

so the powers x^p and x^q are smooth functions of x in the neighborhood under consideration.

Near $(\ell, 0)$, system (32) can therefore be written as

$$Z' = MZ + \mathcal{N}(Z) + d(t)\mathcal{F}(Z),$$

where

$$M = \begin{pmatrix} 0 & 1 \\ -(p-1)B & -A \end{pmatrix},$$

and, writing $Z = (z, y)^T$ with $z = x - \ell$, one may take

$$\mathcal{N}(Z) = \begin{pmatrix} 0 \\ B(\ell+z) - (\ell+z)^p + (p-1)Bz \end{pmatrix}, \qquad \mathcal{F}(Z) = \begin{pmatrix} 0 \\ -(\ell+z)^q \end{pmatrix}.$$

Indeed, since $\ell^{p-1} = B$,

$$B\ell - \ell^p = 0,$$

and

$$\frac{d}{dz}\left[B(\ell+z) - (\ell+z)^p\right]\Big|_{z=0} = -(p-1)B.$$

Therefore Taylor's theorem gives, after possibly decreasing η,

$$|\mathcal{N}(Z)| \le C|Z|^2, \qquad |\mathcal{F}(Z)| \le C$$

for every $|Z| \le \eta$, with a constant $C > 0$ independent of d, T, and T_1.

Since M is Hurwitz, there exist constants $K, \omega > 0$ such that

$$\left\|e^{Mt}\right\| \le Ke^{-\omega t} \qquad (t \ge 0).$$

Here K and ω depend only on the fixed matrix M, hence only on N and p. By the variation-of-constants formula, as long as $|Z(s)| \le \eta$ for $s \in [T, t]$,

$$\begin{aligned} |Z(t)| &\le Ke^{-\omega(t-T)}|Z(T)| \\ &\quad + K\int_T^t e^{-\omega(t-s)}\left(C|Z(s)|^2 + C\delta\right) ds \\ &\le K|Z(T)| + \frac{KC\eta}{\omega}\sup_{T\le s\le t}|Z(s)| + \frac{KC\delta}{\omega}. \end{aligned}$$

Choose $\eta > 0$ sufficiently small that

$$0 < \eta < \frac{\ell}{4} \qquad \text{and} \qquad \frac{KC\eta}{\omega} < \frac{1}{4},$$

and then choose $\eta_0 > 0$ and $\delta > 0$ sufficiently small that

$$K\eta_0 + \frac{KC\delta}{\omega} < \frac{\eta}{2}.$$

We may also assume $\eta_0 < \eta$, so that the initial point already lies strictly inside the neighborhood $\{|Z| < \eta\}$.

Suppose, by contradiction, that there exists a first exit time $t_* \in (T, T_1]$ such that

$$|Z(t_*)| = \eta$$

and

$$|Z(t)| < \eta \qquad (T \le t < t_*).$$

By continuity, $|Z(t)| \le \eta$ on $[T, t_*]$, so the preceding variation-of-constants estimate is valid throughout this interval. Applying the previous estimate on $[T, t_*]$ gives

$$\sup_{T\le t\le t_*} |Z(t)| < \frac{\eta}{2} + \frac{1}{4} \sup_{T\le t\le t_*} |Z(t)|.$$

Hence

$$\sup_{T\le t\le t_*} |Z(t)| < \frac{2\eta}{3} < \eta,$$

contradicting the definition of t_*. Therefore

$$|Z(t)| < \eta \qquad \text{for every } t \in [T, T_1].$$

In particular,

$$x(t) > \ell - \eta > \frac{3\ell}{4} > \frac{\ell}{2},$$

which completes the proof. □

5 Large-amplitude exclusion

For $\varepsilon > 0$, let v_ε denote the maximal positive solution of

$$v_\varepsilon'' + \frac{N-1}{r} v_\varepsilon' + v_\varepsilon^p + \varepsilon v_\varepsilon^q = 0, \qquad v_\varepsilon(0) = 1, \qquad v_\varepsilon'(0) = 0. \tag{33}$$

Let $S(\varepsilon)$ denote its first positive zero, with the convention $S(\varepsilon) = +\infty$ if v_ε remains positive for all $r > 0$.

Lemma 5.1. *Assume that $0 < q < 1 < p$ and $p > p_c$. There exist constants $c > 0$ and $\varepsilon_0 > 0$ such that*

$$S(\varepsilon) \ge c\, \varepsilon^{-\frac{p-1}{2(p-q)}} \tag{34}$$

for every $0 < \varepsilon < \varepsilon_0$.

Proof. Use the Emden–Fowler variables

$$t = \log r, \qquad x_\varepsilon(t) = r^a v_\varepsilon(r), \qquad a = \frac{2}{p-1}.$$

As long as v_ε remains positive, a direct computation gives

$$x_\varepsilon'' + A x_\varepsilon' - B x_\varepsilon + x_\varepsilon^p + \varepsilon e^{mt} x_\varepsilon^q = 0, \tag{35}$$

where

$$m = 2 + a(1-q) = \frac{2(p-q)}{p-1} > 0. \tag{36}$$

By Lemma 4.1, choose T so large that

$$|x_0(T) - \ell| + |x_0'(T)| < \frac{\eta_0}{2},$$

where η_0 is supplied by Lemma 4.2.

Since v_0 is positive on the compact interval $[0, e^T]$, set

$$\kappa_T := \min_{0 \le r \le e^T} v_0(r) > 0.$$

On the region $v \ge \kappa_T/2$, the map

$$(v, \varepsilon) \longmapsto v^p + \varepsilon v^q$$

is locally Lipschitz in v, uniformly for ε in a bounded interval. The radial equation may equivalently be written in integral form as

$$v(r) = 1 - \int_0^r t^{1-N} \int_0^t s^{N-1} \left(v(s)^p + \varepsilon v(s)^q \right) ds\, dt,$$

which removes the apparent singularity at $r = 0$. Since $v_0 \ge \kappa_T > 0$ on $[0, e^T]$, continuous dependence for this integral equation, together with a standard continuation argument inside the region $v \ge \kappa_T/2$, gives

$$v_\varepsilon \ge \frac{\kappa_T}{2} \quad \text{on } [0, e^T], \qquad v_\varepsilon \longrightarrow v_0 \quad \text{in } C^1([0, e^T])$$

as $\varepsilon \to 0^+$. In particular,

$$(x_\varepsilon(T), x_\varepsilon'(T)) \longrightarrow (x_0(T), x_0'(T)) \qquad \text{as } \varepsilon \to 0^+.$$

Hence, after reducing ε_0 if necessary,

$$|x_\varepsilon(T) - \ell| + |x_\varepsilon'(T)| < \eta_0.$$

Choose $\tau_0 < 0$ so that

$$e^{m\tau_0} < \frac{\delta}{2},$$

where δ is given by Lemma 4.2, and define

$$T_\varepsilon := -\frac{1}{m} \log \varepsilon + \tau_0.$$

For $\varepsilon > 0$ sufficiently small,

$$T_\varepsilon > T.$$

Moreover, for every $t \in [T, T_\varepsilon]$,

$$0 \le \varepsilon e^{mt} \le \varepsilon e^{mT_\varepsilon} = e^{m\tau_0} < \frac{\delta}{2} < \delta.$$

If $S(\varepsilon) = +\infty$, there is nothing to prove. We may therefore suppose that $S(\varepsilon) < +\infty$ and set

$$t_\varepsilon^* := \log S(\varepsilon).$$

We claim that x_ε remains positive throughout $[T, T_\varepsilon]$. Suppose, by contradiction, that

$$t_\varepsilon^* \le T_\varepsilon.$$

For every

$$T_1 \in [T, t_\varepsilon^*),$$

the solution is positive on $[T, T_1]$, and hence Lemma 4.2 applies on this interval with

$$d(t) := \varepsilon e^{mt} \in [0, \delta].$$

The lemma gives

$$x_\varepsilon(t) > \frac{\ell}{2}$$

for every $t \in [T, T_1]$. Since $T_1 < t_\varepsilon^*$ is arbitrary,

$$x_\varepsilon(t) \geq \frac{\ell}{2} \qquad \text{for } T \leq t < t_\varepsilon^*.$$

Passing to the limit as $t \uparrow t_\varepsilon^*$ yields

$$x_\varepsilon(t_\varepsilon^*) \geq \frac{\ell}{2} > 0.$$

On the other hand, by the definition of the first zero,

$$x_\varepsilon(t_\varepsilon^*) = S(\varepsilon)^a v_\varepsilon(S(\varepsilon)) = 0,$$

which is a contradiction. Therefore

$$t_\varepsilon^* > T_\varepsilon, \qquad \text{equivalently} \qquad S(\varepsilon) > e^{T_\varepsilon}.$$

Thus, in all cases,

$$S(\varepsilon) > e^{T_\varepsilon}.$$

Finally,

$$e^{T_\varepsilon} = e^{\tau_0} \varepsilon^{-1/m} = e^{\tau_0} \varepsilon^{-\frac{p-1}{2(p-q)}}.$$

Thus (34) holds with

$$c := e^{\tau_0} > 0.$$

□

We now transfer this estimate to the original shooting family.

Proposition 5.2. *Assume that $0 < q < 1 < p$ and $p > p_c$. There exist $A_1 > 0$ and $c_* > 0$ such that $R(A) \geq c_*$ for every $A \geq A_1$.*

Proof. For $A > 0$, set

$$U_A(r) = A v_\varepsilon(s), \qquad s = A^{\frac{p-1}{2}} r, \qquad \varepsilon = A^{q-p}.$$

Then v_ε solves (33), and the corresponding first zeros satisfy

$$R(A) = A^{-\frac{p-1}{2}} S(A^{q-p}). \tag{37}$$

Here the identity is understood in the extended sense: if $S(A^{q-p}) = +\infty$, then $R(A) = +\infty$.

Since $q - p < 0$, we have

$$A^{q-p} \longrightarrow 0 \qquad \text{as } A \to +\infty.$$

Choose $A_1 > 0$ so large that

$$A^{q-p} < \varepsilon_0 \qquad \text{for every } A \geq A_1.$$

Fix $A \geq A_1$. If $S(A^{q-p}) = +\infty$, then $R(A) = +\infty$ and the desired estimate is immediate. We may therefore assume that $S(A^{q-p}) < +\infty$. By Lemma 5.1,

$$\begin{aligned} R(A) &\geq cA^{-\frac{p-1}{2}} \left(A^{q-p}\right)^{-\frac{p-1}{2(p-q)}} \\ &= cA^{-\frac{p-1}{2}} A^{\frac{p-1}{2}} = c. \end{aligned}$$

Indeed, since $q - p = -(p - q)$,

$$\left(A^{q-p}\right)^{-\frac{p-1}{2(p-q)}} = A^{\frac{p-1}{2}}.$$

Thus one may take

$$c_* := c.$$

□

Intermediate amplitudes are easier.

Lemma 5.3. *For every $0 < A_0 < A_1 < \infty$, there exists $c_{0,1} > 0$ such that $R(A) \geq c_{0,1}$ for every $A \in [A_0, A_1]$.*

Proof. Fix $A \in [A_0, A_1]$. If $R(A) = +\infty$, the desired estimate is immediate. Thus suppose that $R(A) < +\infty$. As long as U_A remains positive, it is strictly decreasing, and therefore

$$0 < U_A(r) \leq A \leq A_1.$$

Using (17), for every $0 < r < R(A)$ we obtain

$$-U_A'(r) = r^{1-N} \int_0^r s^{N-1} \left(U_A(s)^q + U_A(s)^p\right) ds \leq \frac{r}{N} \left(A_1^q + A_1^p\right).$$

Integrating from 0 to $r < R(A)$ gives

$$U_A(r) \geq A - \frac{r^2}{2N} \left(A_1^q + A_1^p\right) \geq A_0 - \frac{r^2}{2N} \left(A_1^q + A_1^p\right).$$

For instance, choose

$$c_{0,1} := \left(\frac{NA_0}{A_1^q + A_1^p}\right)^{1/2} > 0.$$

Then

$$\frac{c_{0,1}^2}{2N} \left(A_1^q + A_1^p\right) = \frac{A_0}{2}.$$

We claim that

$$R(A) \geq c_{0,1}.$$

Indeed, if $R(A) < c_{0,1}$, then the previous estimate holds for every $0 < r < R(A)$. Letting $r \to R(A)^-$ and using continuity of U_A, we would obtain

$$0 = U_A(R(A)) \geq A_0 - \frac{R(A)^2}{2N} \left(A_1^q + A_1^p\right) > \frac{A_0}{2},$$

a contradiction. Therefore

$$R(A) \geq c_{0,1},$$

uniformly for $A \in [A_0, A_1]$. □

6 Radial uniqueness for small balls

Theorem 6.1. *There exists $R_* > 0$ such that, for every $0 < R < R_*$, problem (13) admits exactly one positive radial solution, denoted by U_R. Moreover,*

$$\|U_R\|_{L^\infty(B_R)} < A_*.$$

Proof. By Corollary 3.6, choose

$$0 < A_0 < A_*$$

and $R_0 > 0$ such that, for every $0 < R < R_0$, problem (13) admits a positive radial solution whose center value satisfies

$$U(0) < A_0.$$

In particular, by Proposition 3.5, the corresponding shooting parameter tends to zero along the small-amplitude branch as $R \to 0^+$.

Let $A_1 > A_0$ be large enough so that Proposition 5.2 applies. By Lemma 5.3, there exists $c_{0,1} > 0$ such that

$$R(A) \geq c_{0,1} \qquad \text{for every } A \in [A_0, A_1],$$

whereas Proposition 5.2 yields

$$R(A) \geq c_* \qquad \text{for every } A \geq A_1.$$

Consequently,

$$R(A) \geq \min\{c_{0,1}, c_*\} > 0 \qquad \text{for every } A \geq A_0.$$

Choose $R_* > 0$ such that

$$R_* < \min\{R_0, c_{0,1}, c_*\}.$$

Then the small-amplitude existence conclusion holds for every $0 < R < R_*$, while at the same time

$$R_* < \min\{c_{0,1}, c_*\}.$$

Let now $0 < R < R_*$ and let U be any positive radial solution of (13). By the shooting formulation,

$$U(0) = A$$

for some $A > 0$ satisfying

$$R(A) = R.$$

If $A \geq A_0$, then

$$R = R(A) \geq \min\{c_{0,1}, c_*\} > R_*,$$

which contradicts $R < R_*$. Hence

$$U(0) < A_0 < A_*.$$

Since every positive radial shooting profile is strictly decreasing,

$$\|U\|_{L^\infty(B_R)} = U(0) < A_*.$$

Assuming, as in Proposition 3.3, uniqueness in the class of positive solutions under the bound $\|U\|_{L^\infty(B_R)} < A_*$, there is at most one such positive radial solution. Together with the existence of the small-amplitude radial solution, this proves that problem (13) admits exactly one positive radial solution U_R for every $0 < R < R_*$. □

7 From radial uniqueness to full uniqueness

The reaction

$$f_\lambda(s) = \lambda(s^+)^q + (s^+)^p$$

is not locally Lipschitz at $s = 0$, since $0 < q < 1$. Nevertheless, it admits the decomposition

$$f_\lambda = f_1 + f_2, \qquad f_1(s) = (s^+)^p, \qquad f_2(s) = \lambda(s^+)^q,$$

where f_1 is locally Lipschitz and f_2 is nondecreasing.

This is precisely the structural decomposition treated by the moving-plane symmetry theorem of Moreira dos Santos and Nornberg [5]: the zero-order term is the sum of a locally Lipschitz function and a nondecreasing function of the unknown. In the present scalar problem there are no coupling or gradient terms, the operator is the uniformly elliptic Laplacian, the reaction is independent of x (hence in particular nonincreasing in $|x|$), and the domain is the ball. Moreover, f_λ is continuous on $\mathbb{R}$, and the decomposition above satisfies the required structural assumptions on the nonnegative range of the solution. Their theorem therefore applies and shows that every positive classical solution of (1) is radially symmetric about the origin and strictly decreasing in the radial variable.

We can therefore complete the proof of the main result.

Proof of Theorem 1.1. Choose $R_* > 0$ as in Theorem 6.1 and set

$$\lambda_* := R_*^{\frac{2(p-q)}{p-1}}.$$

If

$$0 < \lambda < \lambda_*,$$

then, by (14),

$$R = \lambda^{\frac{p-1}{2(p-q)}} < R_*.$$

By Theorem 6.1, the scaled problem (13) has exactly one positive radial solution. On the other hand, by the symmetry result quoted above, every positive classical solution of (1) is radial. The same symmetry conclusion applies to the scaled problem, since its nonlinearity has the same structural form. Since the scaling (12) gives a one-to-one correspondence between positive solutions of (1) and positive solutions of (13), it follows that (1) admits exactly one positive classical solution, denoted by u_λ.

It remains to determine its size as $\lambda \to 0^+$. Let

$$A = U_R(0).$$

Since

$$R = \lambda^{\frac{p-1}{2(p-q)}} \longrightarrow 0 \qquad \text{as } \lambda \to 0^+,$$

Theorem 6.1 and the exclusion of amplitudes $A \geq A_0$ in its proof imply that

$$A = U_R(0) \longrightarrow 0.$$

Hence Proposition 3.5 is applicable to this family and gives

$$R = \rho_q A^{\frac{1-q}{2}}(1 + o(1)).$$

Therefore

$$A = \rho_q^{-\frac{2}{1-q}} R^{\frac{2}{1-q}}(1 + o(1)).$$

Using

$$R = \lambda^{\frac{p-1}{2(p-q)}},$$

we obtain

$$A = \rho_q^{-\frac{2}{1-q}} \lambda^{\frac{p-1}{(p-q)(1-q)}} \big(1 + o(1)\big).$$

By the scaling (12),

$$u_\lambda(0) = \lambda^{\frac{1}{p-q}} A.$$

Hence

$$\begin{aligned} u_\lambda(0) &= \rho_q^{-\frac{2}{1-q}} \lambda^{\frac{1}{p-q} + \frac{p-1}{(p-q)(1-q)}} \big(1 + o(1)\big) \\ &= \rho_q^{-\frac{2}{1-q}} \lambda^{\frac{1}{1-q}} \big(1 + o(1)\big), \end{aligned}$$

where we used

$$\frac{1}{p-q} + \frac{p-1}{(p-q)(1-q)} = \frac{(1-q)+(p-1)}{(p-q)(1-q)} = \frac{1}{1-q}.$$

Since u_λ is radial and strictly decreasing,

$$\|u_\lambda\|_{L^\infty(B_1)} = u_\lambda(0).$$

Consequently,

$$\|u_\lambda\|_{L^\infty(B_1)} \sim \rho_q^{-\frac{2}{1-q}} \lambda^{\frac{1}{1-q}} \qquad \text{as } \lambda \to 0^+. \tag{38}$$

This completes the proof. □

8 Comments on the mechanism

Remark 8.1. At the Emden–Fowler level, the damping coefficient is

$$A = N - 2 - \frac{4}{p-1}.$$

Thus

$$A > 0 \quad \Longleftrightarrow \quad p > \frac{N+2}{N-2}.$$

In the supercritical regime, we also have $B > 0$, and the linearization at the positive Lane–Emden equilibrium $(\ell, 0)$ has characteristic polynomial

$$\mu^2 + A\mu + (p-1)B.$$

Since

$$A > 0 \qquad \text{and} \qquad (p-1)B > 0,$$

both characteristic roots have strictly negative real parts. Thus $(\ell, 0)$ is exponentially asymptotically stable, and this stability is the key ingredient in the proof of Lemma 5.1.

At the critical exponent,

$$p = \frac{N+2}{N-2},$$

one has $A = 0$, so the positive equilibrium is no longer asymptotically stable by this damping mechanism. Indeed, the linearized equation becomes

$$z'' + (p-1)Bz = 0,$$

whose characteristic roots are

$$\mu_{\pm} = \pm i\sqrt{(p-1)B}.$$

Moreover, the Emden–Fowler equation itself becomes the conservative equation

$$x'' - Bx + x^p = 0,$$

for which the energy

$$E(x, x') = \frac{1}{2}(x')^2 - \frac{B}{2}x^2 + \frac{1}{p+1}x^{p+1}$$

is constant along trajectories. Consequently, the large-amplitude argument used here breaks down. More precisely, the robust trapping mechanism of Lemma 4.2, which relies on exponential asymptotic stability of $(\ell, 0)$, is no longer available when $A = 0$. Thus the critical case requires a different analysis and should not be inferred from the present supercritical argument. This distinction is also compatible with the qualitatively different behavior known for critical concave–convex problems of Ambrosetti–Brezis–Cerami type.

Remark 8.2. The estimate

$$S(\varepsilon) \geq c\varepsilon^{-\frac{p-1}{2(p-q)}}$$

has precisely the scale needed to prevent the original first zero $R(A)$ from collapsing to zero as $A \to \infty$. Indeed, the large-amplitude rescaling gives $R(A) = A^{-\frac{p-1}{2}} S(A^{q-p})$, and therefore

$$\begin{aligned} R(A) &\geq cA^{-\frac{p-1}{2}} \left(A^{q-p}\right)^{-\frac{p-1}{2(p-q)}} \\ &= cA^{-\frac{p-1}{2}} A^{-(q-p)\frac{p-1}{2(p-q)}} \\ &= cA^{-\frac{p-1}{2}} A^{\frac{p-1}{2}} = c, \end{aligned}$$

where we used $q - p = -(p-q)$. Thus the two scaling factors cancel exactly, which is the mechanism behind the uniform lower bound for $R(A)$ at large amplitudes.

Acknowledgment of scope

The uniqueness theorem proved above concerns positive solutions in a ball. The proof uses both the radial shooting geometry of the ball and a radial symmetry theorem for positive solutions. No corresponding uniqueness statement is asserted here for general bounded domains.

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